實驗研究與平台

外部審查提交報告

AI

AI

AI

AI ——

Codex

Claude Code

2026 10 26

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Codex

Claude Code Codex 2026 10 26

AI

AI

[

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Word Times New Roman PDF Iansui

https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/

Drive

https://drive.google.com/drive/folders/1yA7Vx4kZUW0Im2B24o4K813Wxh6N5BlY

http://100.83.58.39:8771/ http://100.83.58.39:8771/architecture

Tailscale AI

1 □□□□□□□□□

1.1

AI

47 AI ——

1.2

NotebookLM

A Ctrl F BM25 B

AI

corpus.json

NotebookLM A

Google

1.3

RQ1 A B

RQ2 B

RQ1 RQ2

2 查核摘要與證據層級

2.1

commit 62a37d9f213e4057929d31352d96f1bc4007f3fe 14a9118 app static/app.js A B 8bb7bb7 34 1 35

Engine 3 A 6 B B 10 development-probes.json

1

主張	證據與判定
A	Engine.literal str.find 0 41
B	BGE revision 768 API token
B	B-01 B-02
B	B-01
	ID
	50 512 token

2.2

AI

3 來源資料庫與資料品質

3.1

1,781 1,781 tarshar-vault commit 1dd75bfd4dcd06039f1803f2a65b95c9c64c464d projects/ /thesis-platform/data/corpus.json 1,772 9 SHA256 platform/data/manifest.json ZIP SHA256 source-manifest.json

built 2026-09-21 purple_revision 2026-10-22-development-2 built 2025-11-21 2026-10-22

2

類別或檢查	本次回讀結果
	1,481
	157
	5
	138
	1,781 35
	157
ID	0
	1,757
	1,029
1,500	0
512 tokens	50 1,066 tokens tokens

3.2

https://law.moj.gov.tw/
https://laws.mol.gov.tw/ https://45plus.wda.gov.tw/ HTTPS

source-catalog.csv 35 corpus-inventory.csv ID token data/corpus.json HTML JSON

2 https://laws.mol.gov.tw/FLAW/FLAWDOC01.aspx?id=FL015153&flno=2 https://laws.mol.gov.tw/FLAW/FLAWDAT0202.aspx?id=FL015153 HTML

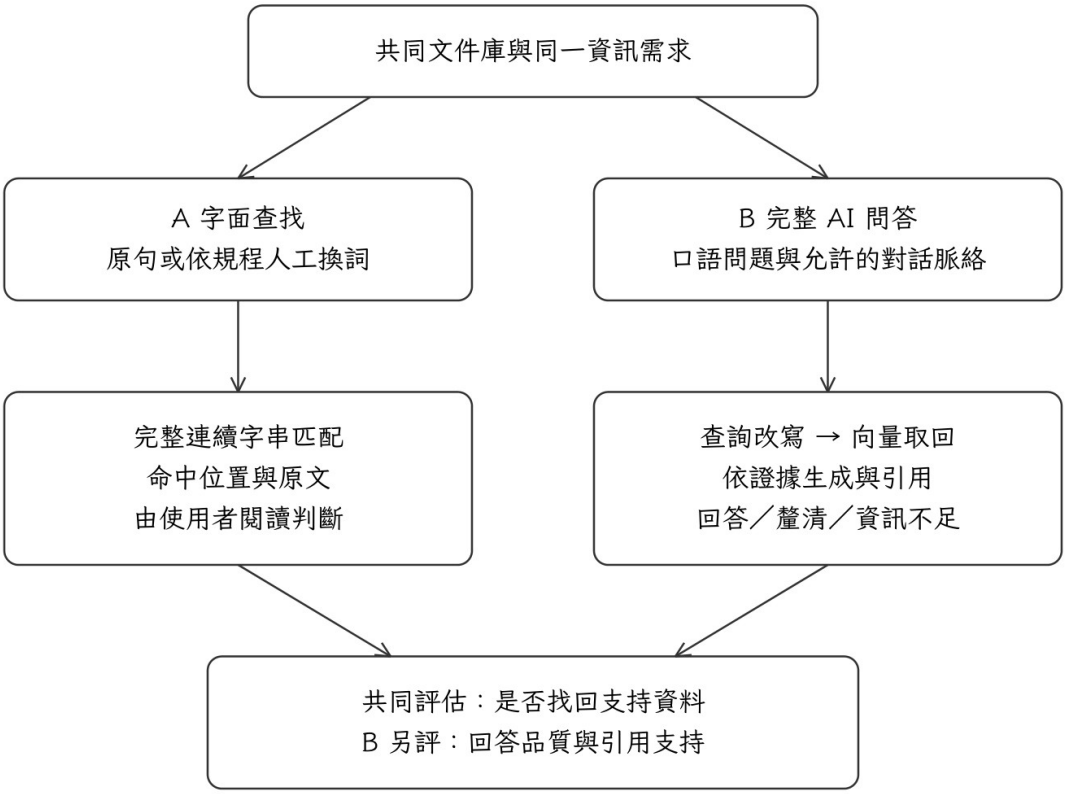
157
import_parental_regulation.py

3.3

tokenizer revision max_seq_length 512 50
50

1,500 1,029
token

A B B



4 平台架構與實際運作

4.1

正式題組、操作規程與評分尚待確認

1

server

query provider API

3

檔案或函式	內容
app/engine.py	A B ID
app/provider.py	OpenAI REST JSON
app/server.py	HTTP API B CSV
static/app.js	150

□□□□□	□□
data/corpus.json	
requirements.lock.txt	Python
tests/test_engine.py test_server.py	
scripts/ import_parental_regulation.py	HTML

4.2 A

query law no text

strip Ctrl F

corpus 20 offset 20

A A

4.3 B

BGE

2

4.4

12 2,000

ID ID ID

insufficient trace

citations hits

5 模型設定與提示詞可稽核性

4

項目	目前值
	BAAI/bge-base-zh-v1.5
revision	f03589ceff5aac7111bd60cfc7d497ca17ecac65
	768 512 tokens
	gpt-5.4-mini-2026-03-17 API
API	https://api.openai.com/v1/chat/completions
	temperature=0 response_format=json_object seed
	gpt-5.4 reasoning_effort=none
	600 1,800 max_completion_tokens
	500 12 2,000
API	50,000 token
	60 timeout HTTP transport retries=0
	SQLite 100

temperature=0

REWRITE_SYSTEM ANSWER_SYSTEM QUERY_INSTRUCTION prompts.json .md AI API trace JSON

Claude Code B Claude OpenAI API B Claude CLI USD 2.268408

6 □□□□□□□□□

6.1

https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/platform-source.zip https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/data/corpus.json repository https://github.com/tarshar4242/thesis-semantic-platform ZIP

ZIP 89 .env runtime Git API

```
unzip platform-source.zip
cd platform
python3.11 -m venv .venv
.venv/bin/python -m pip install -r requirements.lock.txt
cp .env.example .env
chmod 600 .env
```

PDF submission.md PDF

.env OPENAI_API_KEY LLM_MODEL

Git A B Mac Windows Linux

6.2

```
#        revision
.venv/bin/python -c 'from sentence_transformers import SentenceTransformer;
SentenceTransformer("BAAI/bge-base-zh-v1.5",
revision="f03589ceff5aac7111bd60cfc7d497ca17ecac65")'
.venv/bin/python -m pytest -q
HF_HUB_OFFLINE=1 .venv/bin/python -m uvicorn app.server:app --host 127.0.0.1
--port 8771 --workers 1 --no-access-log
```

http://127.0.0.1:8771/ B

Mac mini LaunchAgent 100.83.58.39:8771

Tailscale Funnel

6.3 API

```
curl -sS http://127.0.0.1:8771/api/status
curl -sS http://127.0.0.1:8771/api/query \
-H 'Content-Type: application/json' \
-d '{"group":"A","query":"  ","history":[],"session":"review-
a","task":"dev-001","kind":"development","offset":0}'
curl -sS http://127.0.0.1:8771/api/query \
-H 'Content-Type: application/json' \
-d '{"group":"B","query":"              ","history":[],"session":"review-
b","task":"dev-001","kind":"development"}'
```

/api/status /api/document/{id} /api/export CSV API kind development smoke formal results/demo_log/YYYYMMDD.jsonl request result raw_text evidence API raw_text parsed evidence B 429 422 query

6.4

audit-corpus.py run-development-probes.py platform-source.zip tokenizer API probe-output

```
HF_HUB_OFFLINE=1 platform/.venv/bin/python audit-corpus.py platform
HF_HUB_OFFLINE=1 platform/.venv/bin/python run-development-probes.py
platform
```

6.5

A revision B B

7 研究方法與公平操作建議

7.1

API

7.2

doc_id

AI

development-probes.json corpus

7.3

S1 B history A

S2 A B S2 S1 A B

[3 A 1 20 3 20]

7.4 A

LLM A

gold

action_type=page offset

7.5 B

history

7.6

A B session AB BA

A B

B

A

8 評分規準與分析計畫

8.1 B

5

層次	A	B
		20
		trace.evidence 1,500
		citations
		points answer

m m

recall MAP

B

ID A

8.2

trace insufficient

A B B A

attempt retry_of HTTP

8.3

ID

B

AI

8.4

B

A B McNemar

B 3

9 本輪實跑與已知失敗

6

編號	問題類型	程式回應與限制
A-01		0
A-02		41 41
A-03		38
B-01		answered 20 3
B-02		answered 20 2
B-03		clarify
B-04		insufficient 16
B-05		clarify
B-06	B-01	answered 17 2

Engine B-02 static/app.js history

9.1

B-01 30 2 30
ID

B-06 20 17 3 2 30
temperature=0

9.2

ID _validate_answer
answered citation-validator-boundary.json
ID

9.3

35 pytest ID

FakeLLM mock transport

10 修正清單與正式實驗啟動條件

7

優先	缺口	最小可查驗完成條件
	50	
	157	
	CODE_HASH	manifest
	CSV	
		DB

100 53 63

DB CODE_HASH app/*.py API

11 □□□□□□□□

11.1

1.
2. A B
3. source-manifest corpus-inventory URL token ID
4.
5. API evidence B-01 30
6.
7.
8. DOI
9. Tailscale
10. ID

11.2

B

A B

12 □□□□□□□

2026-10-26 DOI

references.json .md

12.1 R01 Belkin, N. J

Belkin, N. J. (1980). Anomalous states of knowledge as a basis for information retrieval. Canadian Journal of Information Science, 5, 133–143.

https://wp.comminfo.rutgers.edu/belkin/wp-content/uploads/sites/75/2023/03/publications-2023-03.pdf

DOI

12.2 R02 Blair, D. C., & Maron, M. E

Blair, D. C., & Maron, M. E. (1985). An evaluation of retrieval effectiveness for a full-text document-retrieval system. Communications of the ACM, 28(3), 289–299.

https://doi.org/10.1145/3166.3197

https://dl.acm.org/doi/10.1145/3166.3197

1985 Google Ctrl F

12.3 R03 Es, S., James, J., Espinosa Anke, L., & Schockaert, S

Es, S., James, J., Espinosa Anke, L., & Schockaert, S. (2024). RAGAs: Automated evaluation of retrieval augmented generation. In Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations (pp. 150–158). Association for Computational Linguistics.

https://doi.org/10.18653/v1/2024.eacl-demo.16

https://aclanthology.org/2024.eacl-demo.16/

12.4 R04 Furnas, G. W., Landauer, T. K., Gomez, L. M., & Dumais, S. T

Furnas, G. W., Landauer, T. K., Gomez, L. M., & Dumais, S. T. (1987). The vocabulary problem in human-system communication. Communications of the ACM, 30(11), 964–971.

https://doi.org/10.1145/32206.32212

https://dl.acm.org/doi/10.1145/32206.32212

A

12.5 R05 Gao, T., Yen, H., Yu, J., & Chen, D

Gao, T., Yen, H., Yu, J., & Chen, D. (2023). Enabling large language models to generate text with citations. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 6465–6488). Association for Computational Linguistics.

https://doi.org/10.18653/v1/2023.emnlp-main.398

https://aclanthology.org/2023.emnlp-main.398/

ALCE

ID ALCE

12.6 R06 Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W

Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W. (2020). Dense passage retrieval for open-domain question answering. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP) (pp. 6769–6781). Association for Computational Linguistics.

https://doi.org/10.18653/v1/2020.emnlp-main.550

https://aclanthology.org/2020.emnlp-main.550/

DPR BGE DPR BM25 BGE Ctrl F

12.7 R07 Kelly, D

Kelly, D. (2009). Methods for evaluating interactive information retrieval systems with users. Foundations and Trends in Information Retrieval, 3(1–2), 1–224.

https://doi.org/10.1561/1500000012

https://www.nowpublishers.com/article/Details/INR-012

224

12.8 R08 Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks [Preprint; accepted at NeurIPS 2020]. arXiv.

https://doi.org/10.48550/arXiv.2005.11401

https://arxiv.org/abs/2005.11401

RAG

BGE API

12.9 R09 Nonaka, I

Nonaka, I. (1994). A dynamic theory of organizational knowledge creation. Organization Science, 5(1), 14–37.

https://doi.org/10.1287/orsc.5.1.14

https://pubsonline.informs.org/toc/orsc/5/1

12.10 R10 Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., & Nie, J.-Y

Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., & Nie, J.-Y. (2024). C-Pack: Packed resources for general Chinese embeddings. In Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (pp. 641–649). Association for Computing Machinery.

https://doi.org/10.1145/3626772.3657878

https://api.crossref.org/works/10.1145/3626772.3657878

BGE

12.11 T01 Beijing Academy of Artificial Intelligence

Beijing Academy of Artificial Intelligence. (n.d.). BAAI/bge-base-zh-v1.5 [Model card and configuration; revision f03589ceff5aac7111bd60cfc7d497ca17ecac65]. Hugging Face.

https://huggingface.co/BAAI/bge-base-zh-v1.5/blob/f03589ceff5aac7111bd60cfc7d497ca17ecac65/README.md

768 max_seq_length=512

model card 512 tokenizer token 512

reranker reranker

12.12 T02 Google

Google. (n.d.). Frequently asked questions. Gemini Notebook Help. Retrieved October 26, 2026.

https://support.google.com/gemininotebook/answer/16269187?hl=en

FAQ

12.13 T03 Google

Google. (n.d.). Learn about Gemini Notebook. Gemini Notebook Help. Retrieved October 26, 2026.

https://support.google.com/gemininotebook/answer/16164461?hl=en

13 完整附件與版本追溯

13.1

8

檔案	供評審使用的內容
submission.pdf submission.docx	
submission.md	UTF-8
ai-review-input.md	AI
platform-source.zip	89
submission-bundle.zip	
source-manifest.json	commit 89 SHA256
source-catalog.csv	35
evidence/corpus-inventory.csv	1,781 ID URL
prompts.json	SHA256
evidence/development-probes.json	3 A 6 B
evidence/citation-validator-boundary.json	ID
references.json	DOI
protocol-template.json grading-template.csv	
.md	Claude Code
evidence/ environment.json	

□□	□□□□□□□□
SHA256SUMS.txt	

13.2

https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/code.md

https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/prompts.json

AI https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/ai-review-input.md

https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/submission/references.json

47 https://tarshar4242.github.io/learn-ai-with-tarshar/thesis-library/purple/materials/thesis.pdf

13.3

Codex

A gold AI A

A CSV literal CSV